



Topographic Wetness Index and Terrain Ruggedness Index in geomorphic characterisation of landslide terrains, on examples from the Sudetes, SW Poland

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with 7 figures and 2 tables

Abstract. Topographic Wetness Index (TWI) and Terrain Ruggedness Index (TRI) are secondary geomorphometric parameters used to describe and quantify local relief. In this paper their usefulness in geomorphological studies of landslides is explored. Using a sample of twelve objects from the Sudetes, SW Poland, spatial and statistical distributions of TWI and TRI values are presented and discussed. Both TWI and TRI prove capable of differentiating landslide population into smaller groups, consistent with their variable origin and mechanism of displacement. Considerable disruption of original slope surface during movement results in high TWI range values and low TRI values, whereas for translational rock-slides and slumps the relationship is opposite. Spatial patterns of high TWI values help to identify lines of preferential drainage of a landslide body. Directional analysis shows that different types of displacement have their specific morphological signatures. Both TWI and TRI provide new information about landslide morphology and are recommended for use, especially to characterize depositional parts of landslides. The study is based on high resolution digital elevation model derived from airborne laser scanning (LiDAR) which proved useful in detailed characterization of landslide morphology.

Keywords: Geomorphometry, surface roughness, Topographic Wetness Index, Terrain Ruggedness Index, landslides, Sudetes

1 Introduction

Geomorphometric approach to analyse landforms produced by landslides has considerable potential to enhance our understanding of the phenomenon of landsliding and owing to increasing availability of high-resolution elevation data, such as those derived from airborne laser scanning of Earth surface (LiDAR = Light Detection and Ranging), it has become a powerful tool in landslide research. The usefulness of geomorphometry in landslide studies is at least twofold. First, landslides and related phenomena can be characterized in a more objective way using quantitative indices, which also provide solid background for any further studies, including comparative analysis and landslide susceptibility assessment. Second, geomorphometry may reveal 'hidden' topographic features of landslide terrains, otherwise unaccounted for. In this way, it offers insights into the origin and subsequent behaviour of individual landslides.

Geomorphometry has been used in various studies, aimed at different targets. Perhaps the most common is the use of metrics of landslide terrains to aid automatic recognition of landslide-affected slopes. An underlying assumption holds that remodelling by mass movements markedly changes slope surface characteristics and these can be detected by quantitative, objec-

tive DEM (digital elevation model) analysis. This approach is particularly attractive for large areas, where systematic visual interpretation of DEM is hardly possible. Thus, algorithms and procedures which would help to recognize landslide-affected terrains (SEKIGUCHI & SATO 2004, ARDIZZONE et al. 2007, LIU et al. 2009, PALENZUELA et al. 2015) and to delimit terrain units most prone to future landsliding (STOLZ & HUGGEL 2008, PETSCHKO et al. 2016) are sought. A common premise in these studies is that sliding and then deposition of the failed mass, often disintegrated en route, will result in considerable increase of local topographic complexity in respect to adjacent, unaffected slope (e.g. McKEAN & ROERING 2004, BOOTH et al. 2009). However, LIU et al. (2009) followed different logic, that the surface exposed by the slide would have smoother topography and hence, focused on terrain patches less rough than the surrounding ones in the Alishan area of Taiwan. Thus, depletion areas may have different topographic signatures than depositional areas.

Computing and analysing measures of surface (topographic) roughness is thus part of geomorphometric approach and there are two main reasons why roughness is focused on. First, it is hoped that geomorphometric parameters will help to differentiate between landslides of different types as different types of displacements are expected to produce dissimilar topographic signatures. Second, it is assumed that geomorphic evolution of a landslide body through time will result in smoothing its surface through diffusive mechanism of slope degradation and local deposition, however varied it may have been initially. Thus, landslides of the same type and material involved but different in terms of age should show inverse relationship between surface roughness and age. Following these assumptions, LAHUSEN et al. (2016) have recently analysed a sample of 25 deep-seated landslides in glacial sediments in Washington, USA, and used a simple measure of their surface roughness – average standard deviation of slope in a 15×15 m moving window – as a proxy of age.

Literature review shows that various indices of roughness have been applied to characterize landslide terrains. They have been recently reviewed by BERTI et al. (2013) who evaluated 10 different measures, seeking ones which suit best to discriminate active landslides and predict the presence of landslides while carrying out automatic mapping. They concluded that simple measures such as root mean square of height or slope, or standard deviation of elevation or slope, perform equally well as the more complex ones. McKEAN & ROERING (2004) accounted for variability in slope and aspect at the same time, applying unit orientation vectors and analysing their statistical variability using eigenvalue ratios. They particularly recommended this technique for flattish terrain, where changes in local aspect are the greatest manifestations of roughness. Standard deviation of height above a baseline surface as a single measure has been used by GLENN et al. (2006) and LAHUSEN et al. (2016), although the former authors were interested in very local diversity of landslide surface, adopting only 5×5 cell size. Different ruggedness indices were used by VAN DEN ECKHAUT et al. (2012) and CROSTA et al. (2013), to assist landslide delimitation using object-oriented identification and analysis of variables controlling deep-seated gravitational slope deformations in the Alps, respectively. In the former study, Terrain Ruggedness Index by RILEY et al. (1999) was used.

In this study we focus on a sample of 12 landslides from the Sudetes, SW Poland, fairly well recognized in terms of morphological diversity, although at a rather descriptive level, and

subject them to geomorphometric analysis, with the main focus on two measures: Topographic Wetness Index (TWI) (MOORE et al. 1991, SØRENSEN et al. 2006, GRUBER & PECKHAM 2009) and Terrain Ruggedness Index (TRI) (RILEY et al. 1999). Whereas the latter has been used in the past, we were unable to find application of the former to characterize landslide terrains. Topographic Wetness Index is a tool to indicate areas accumulating water flow, often with seasonally and permanently waterlogged ground. As such, it is very useful to show geomorphic complexity of a landslide terrain, including the pattern of topographic highs ('dry' areas) and lows ('wet' areas). Terrain Ruggedness Index is a parameter that quantifies surface roughness through consideration of absolute elevations in the surrounding of a given raster cell.

In our research we intend to answer two questions relevant to the use of these indices. First, do these measures provide new information about landslide morphology and if so, what kind of information? Second, are these indices helpful to discriminate between landslides of different origin? Concerning TWI, we intend to investigate whether visual analysis of TWI spatial distribution can be supplemented by a more objective, quantitative approach to landslide terrain and if it is possible to use TWI to distinguish between landslides of different types. Likewise, we look at TRI from this perspective. In other words, do these indices reveal distinctive topographic (geomorphometric) signatures of different landslides? Thus, this paper is primarily a methodical experiment, aimed at critical evaluation of two indices of topographic complexity, rather than a systematic geomorphometric study of landslides from a region.

2 Study area and landslide description

2.1 *Landslides in the Sudetes*

Landslides selected for this study are located in different parts of the Sudetes Mountains, in SW Poland (Fig. 1). The Sudetes are a medium-high mountain range (maximum altitude 1602 m a.s.l., typically 600–1000 m a.s.l.) and belong to the Central European discontinuous belt of rejuvenated Variscan orogenic terrains, subject to differential vertical movements in the Cenozoic. Slopes are locally fairly steep ($> 25^\circ$), especially along fault-generated escarpments, in deeply incised river valleys, and at some lithological boundaries, where rocks of contrasting strength juxtapose. These settings favour slope failures, particularly if high topographic gradients are coupled with lithological and structural predisposition. Conditions for landsliding include the presence of hard and permeable sedimentary or volcanic rocks on much weaker sedimentary materials, e.g. sandstone over mudstone as in the Stołowe Mountains (MIGOŃ & KASPRZAK 2016) or trachyandesites, rhyolites or welded tuffs over shales as in the Kamienne Mountains (MIGOŃ et al. 2014a). In other locations ancient fault planes and lithological boundary surfaces may act as sliding surfaces. Fluvial undercutting of slopes has been responsible for slope failures on outer meander bends, e.g. in Bardo (CIĘŻKOWSKI & KOSZELA 1988, MIGOŃ et al. 2002). Most landslides are relict (prehistoric), with only a few events reported in historical sources, e.g. in Bardo in 1598. However, whether they date back to the Pleistocene, Pleistocene/Holocene transition, or to certain periods of the Holocene, remains largely unknown.

Landslides in the Sudetes represent different types recognized in landslide classifications (VARNES 1978, DIKAU et al. 1996, HUNGR et al. 2014). Among these various types of displace-

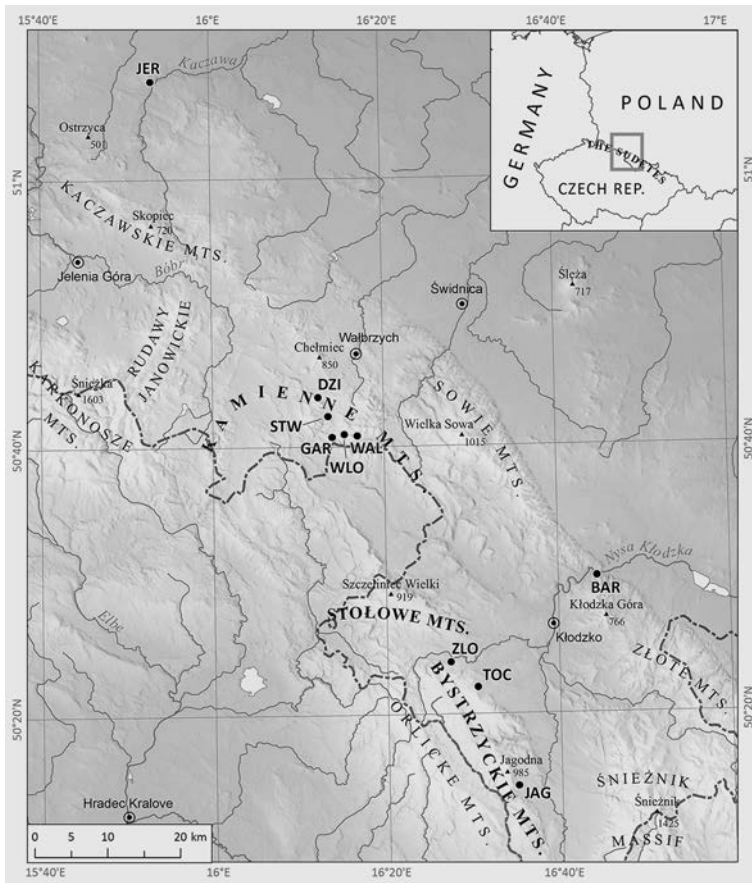


Fig. 1. Location of landslides analyzed in the paper. Letter codes of landslide names as in Table 1. Note that TOC (standing for Toczek) includes three different landslides located next to one another.

ments are translational slides, rotational slides with shear surfaces at different depths, mud-flows and debris flows, topples, and lateral spreads. Landform inventories suggest that many slides are complex and initial slides transformed into flows in the lower slope, either extrusive or channelized, depending on local relief (MIGOŃ et al. 2014a, RÓŻYCKA et al. 2015). Likewise, they vary in areal coverage, from localized displacements extending over 1–2 ha to fairly large slope instabilities, affecting 20 ha and more.

2.2 Characteristics of landslide localities

Twelve landslides selected for this study and subject to geomorphometric analysis are meant to represent a variety of morphological and morphogenetic types (Table 1, Figs. 2 and 3). Five of them are located in the Bystrzyckie Mountains in the Middle Sudetes, including one large landslide complex at Mt Toczek, divided into three separate landslide bodies, each with its own distinctive morphology (RÓŻYCKA et al. 2015, MIGOŃ et al. 2016b). Toczek E landslide, with the

Table 1. Morphometric and geomorphic features of landslides considered in this study.

Name	Code	Area (ha)	Length (m)	Maximum width (m)	L/W ratio	Height range (m)	Mean slope (°)	Lithology	Type of landslide	Morphology
Jagodna	JAG	28.3	928	476	1.9	144	10.0	Calcareous mudstones	Flow	Indistinct head scarp, hummocky topography of landslide body, minor slope-perpendicular steps; spread over the slope
Toczek S	TOS	13.6	755	257	2.9	147	13.8	Sandstone on mudstone and marl	Flow	Valley-confined, with hummocky topography of the tongue; indistinct head scarp
Toczek E	TOE	42.8	753	768	1.0	184	13.9	Sandstone on mudstone and marl	Complex	Steep head scarp backed by belt of ongoing deformations, with trenches and open fissures; series of displaced blocks below; hummocky topography in the lower slope, distinct toe
Toczek N	TON	9.6	307	387	0.8	118	19.6	Sandstone on mudstone and marl	Rotational	Steep head scarp with bedrock partially exposed; series of displaced blocks below, suggesting multiple rotational slide; the toe probably partly fluvially undercut
Złota	ZLO	4.9	204	318	0.6	93	22.4	Sandstone on clayed marls and calcareous claystones	Rotational	Simple rotational slide, with one coherent displaced block; steep head scarp but no continuous bedrock outcrop
Dzikowiec	DZI	12.3	285	603	0.5	140	26.0	Trachyan-desite on shale	Rotational	Series of several rotational slumps, next to one another; indistinct head scarp
Stożek Wielki	STW	14.1	569	379	1.5	231	27.3	Trachyan-desite on shale	Translational	Translational slide of wedge failure type; head scarps partly covered by talus; displaced block disintegrated and dissected
Garbatka	GAR	21.9	1046	211	5.0	222	14.0	Rhyolitic tuff	Flow	Shallow slides in two adjacent slope hollows transformed into flow travelling to the valley floor; pre-Holocene age of landslide and very subdued morphology
Włostowa	WLO	23.0	664	499	1.3	236	21.6	Rhyolite on shale	Complex	Multiple rotational slide (west) and rockslide (east) in the upper part; significant surface deformation in the lower part, hence hummocky topography
Waligóra	WAL	11.8	402	316	1.3	203	21.9	Rhyolite on shale	Complex	Rotational slump in the upper part, transformed into flow-like movement downslope; landslide body split and directed into two adjacent valleys; extensive talus below head scarp
Bardo	BAR	4.3	342	186	1.8	184	31.0	Greywacke, shale	Translational	Deep landslide scar and up to 20 m high bedrock head scarp, with extensive talus below; lower part in unconfined valley bottom with hummocky topography and big boulders
Jerzmanice	JER	3.5	180	287	0.6	78	25.9	Slate	Rotational	Simple rotational slump, stepped slope profile, arcuate head scarp, toe partially eroded by the river

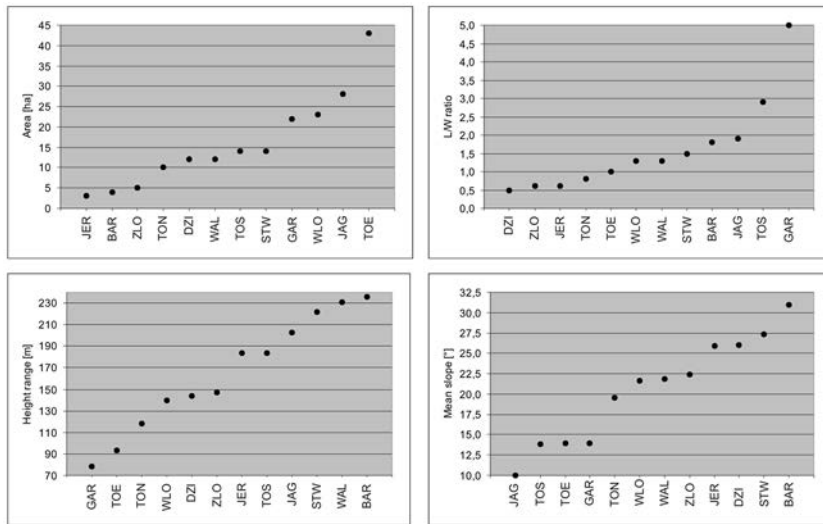


Fig. 2. Plots to show diversity of landslides considered in this paper in terms of basic morphometric parameters presented in an increasing order of numerical values. Letter codes of landslide names as in Table 1.

total area of 43 ha, is the largest single landslide recognized in the Polish part of the Sudetes. The other two (Jagodna, Złota) have not been subject to detailed studies yet. Another five examples have been selected from a large population present in the Kamienne Mountains, where nearly 50 individual landslides have been identified (MIGOŃ et al. 2014a). Criteria of selection included large size (minimum 10 ha) and morphogenetic diversity. Thus, translational, rotational and flow slides are represented, whilst two other are complex landslides, where single or multiple slid blocks in the upper slope give way to flow in the mid- and lower slope. Morphology, activity and probable age of some of these landslide terrains (Garbatka, Stożek Wielki, Waligóra and Włostowa) have already been discussed elsewhere (MIGOŃ et al. 2010, 2014b, 2016a). The sample is supplemented by two landslides from other parts of the Sudetes. The 1598 rockslide in Bardo is the oldest large-scale slope failure in the Sudetes documented in historical sources (CIEŻKOWSKI & KOSZELA 1988), whereas the landslide in Jerzmanice-Zdrój, although rather small (3.5 ha), is a very good, 'pure' example of a rotational slump. The latter two landslides occurred on outer meander bends of Nysa Kłodzka and Kaczawa rivers, respectively. Their toe parts were located in the valley floors and subsequently partly eroded, hence straight outlines of landslide bodies in the distal positions. However, the width of valley floors – accommodation space for slid masses – suggests that eroded distal parts were rather short, a few tens of metres long. Landslides selected for this study are diverse in terms of area (3–46 ha, with the majority being of medium size 10–25 ha), height range (from less than 80 m to more than 230 m), elongation and mean slope (Fig. 2), which ensures fairly good representation. None of these landslides has been reported to show activity in the last 200 years or so.

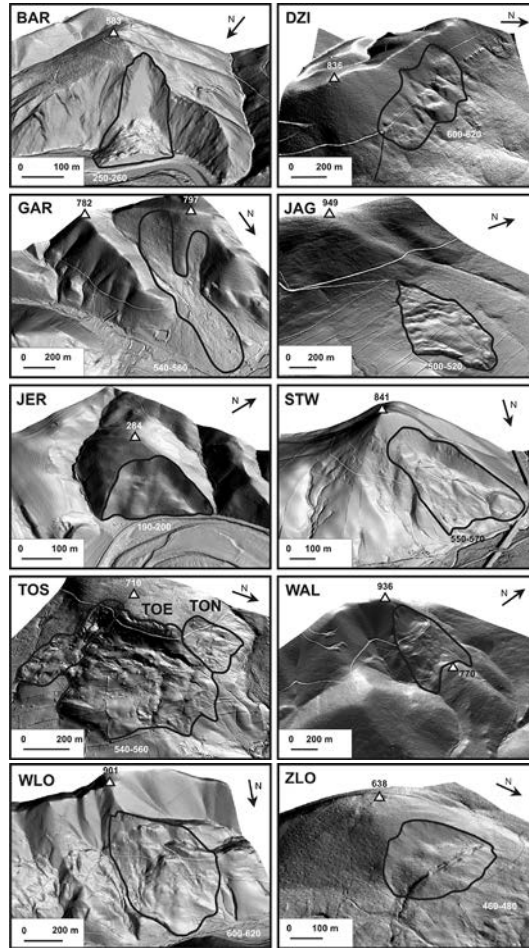


Fig. 3. Three-dimensional terrain models showing the morphology of studied landslides.

3 Data sources and analytical procedures

3.1 Digital elevation model and software

Landslide geomorphometry was carried out on digital elevation model of 1×1 m spatial resolution built from primary data acquired by high-resolution airborne LiDAR (Light Detection and Ranging). The relevant segmented dataset was purchased from the Polish Centre of Geodetic and Cartographic Documentation. Laser scanning was performed with the accuracy of 4–6 points per 1 m^2 , the mean elevation error is 0.05–0.15 m, and the maximal error does not exceed 0.3 m. The mean localization error of the ALS point cloud is less than 0.5 m (Wężyk 2014). The final DEM, produced by removal of points reflected from terrain cover elements (vegetation, infrastructure), is a bare ground type model and offers the best approximation of elevation dis-

tribution in the study area. Boundaries of landslides were delineated using combination of DEM analysis and field geomorphological mapping.

Digital elevation data were processed and visualized using SAGA GIS and ArcGIS software. The former software was used to generate maps of spatial distribution of TWI and TRI values and in reclassification procedures, whereas in the latter images of shaded relief and rose diagrams of drainage directions identified from reclassified TWI maps were produced.

3.2 *Topographic Wetness Index (TWI)*

Topographic Wetness Index, defined by the following equation: $TWI = \ln(A/\tan\beta)$, where A is the upslope contributing area and $\tan\beta$ is the local slope (MOORE et al. 1991, SØRENSEN et al. 2006, GRUBER & PECKHAM 2009), was used in two ways. First, to characterize spatial topographic diversity and complexity of landslide terrains and second, to characterize landslides quantitatively by focusing on TWI range within their boundaries.

In the former, maps of TWI spatial distribution were generated and analyzed visually, to identify patterns of preferential drainage of landslide areas as shown by raster cells of high index values. However, it was noted that forest roads often emerge as thin lines of high TWI values and these were identified and ignored. Then, reclassification was carried out in order to distinguish raster cells with the highest TWI values, understood as those above mean value plus two standard deviations. Most of the very high TWI values clustered as narrow elongated areas and these linear zones, except forestry roads, were vectorised and divided into smaller segments of identical orientation. Length and azimuth of these segments in downslope direction constituted a database from which rose diagrams to show frequency of drainage directions in 10° intervals were generated. For four landslides of clearly bipartite morphology three diagrams were generated: for the entire area and for each sub-area.

TWI range was considered in three variants: (a) simple max–min difference within the entire dataset for a landslide, which is an absolute range value; (b) difference between the mean value for a landslide increased by one standard deviation and the mean value decreased by one standard deviation (68 per cent of all raster cells); (c) difference between the mean value for a landslide increased by two standard deviations and the mean value decreased by two standard deviations (95 per cent of all raster cells).

3.3 *Terrain Ruggedness Index (TRI)*

This parameter is computed in the following way: from elevation of a central cell, for which the TRI value is calculated, elevation values of neighbouring cells are subtracted. The resultant differences are multiplied by themselves, summed up and squared once again (RILEY et al. 1999), possibly to highlight morphological heterogeneity. TRI is expressed as $(\sum(z_c - z_i)^2)^2$, where z_c is the elevation of a central cell and z_i is the elevation of one of the eight neighbouring cells ($i = 1, 2, \dots, 8$).

The computation algorithm of TRI is sensitive to local differences in elevation and hence, the presence of a long and steep head scarp is likely to increase the average value of TRI calculated for each landslide. To check its influence, TRI average values were calculated in three different variants: (a) for an entire landslide; (b) without the head scarp in its exposed part; (c)

for the area of primary deposition (landslide body), i.e. without the head scarp and talus slope below. In one case delimitation of head scarp was not possible (Toczek S), whereas in another one talus slope could not be unequivocally distinguished (Jagodna). These variants served the purpose to provide data for more accurate characterization of landslide depositional bodies using the mean value.

Finally, spatial distribution of raster cells with lowest TRI values was analysed. Two variants have been tested, focusing on the distribution of raster cells whose values were in the lowermost one fifth of the min–max range and in the lowermost one tenth of the min–max range. They were separated from the datasets involving depositional zones (i.e. without head scarps and talus), their percentage calculated, and distribution shown on maps for each landslide.

4 Results

4.1 *Landslide topography revealed by Topographic Wetness Index*

4.1.1 *Spatial patterns*

Topographic Wetness Index reveals diversity and complexity of landslide topographic surface, highlighting zones of preferential drainage of the landslide body, as well as relatively dry areas within its boundaries. The upper parts of landslides, especially steep head scarps (e.g. Bardo, Waligóra, Toczek N, Jerzmanice, Złota) invariably show low TWI values. The association of the depletion zone with low TWI is less evident for landslides generated in slope hollows and initiated as shallow slides (Garbatka, Toczek S). The middle and lower parts are clearly differentiated into rather irregular zones of low to moderate TWI values, denoting local elevations and steps, and linear areas of high TWI (Fig. 4). Ignoring anthropic elements (roads), the latter indicate elongated low-lying areas which collect water and route it beyond the landslide area. These areas vary in terms of width and are non-uniformly distributed across the landslide body. Two contrasting cases are illustrated by landslides of Jagodna and Waligóra. In the former, a system of broadly parallel but rather narrow drainage lines is revealed, with minor drainage lines converging into major ones, so that eventually only two drainage lines leave the landslide body. In the latter, instead, a broad ‘wet’ zone in the southern part of the landslide can be recognized. The distribution of high TWI values shows different spatial patterns too. A simple pattern is illustrated by Jerzmanice landslide where two drainage lines follow directions of lateral scarps, whereas a far more complicated one typifies Włostowa landslide, where preferential flow lines tend to be perpendicular to the general slope in the upper part and parallel to it in the lower part.

Recognizing that automatically-generated TWI maps displayed with linear scale may be misleading since they do not differentiate raster cells with the highest values, reclassified images have been prepared. They show spatial distribution of raster cells with the highest values of TWI for each landslide, which exceed the mean value increased by two standard deviations and hence, are contained within 5 per cent of cells with extreme values (Figs. 5 and 6), making the studied landslides fully suited to comparative analysis.

The spatial patterns consist of continuous and discontinuous lines/zones of high TWI and vary from simple ones, with one or two main directions present, to much more complicated

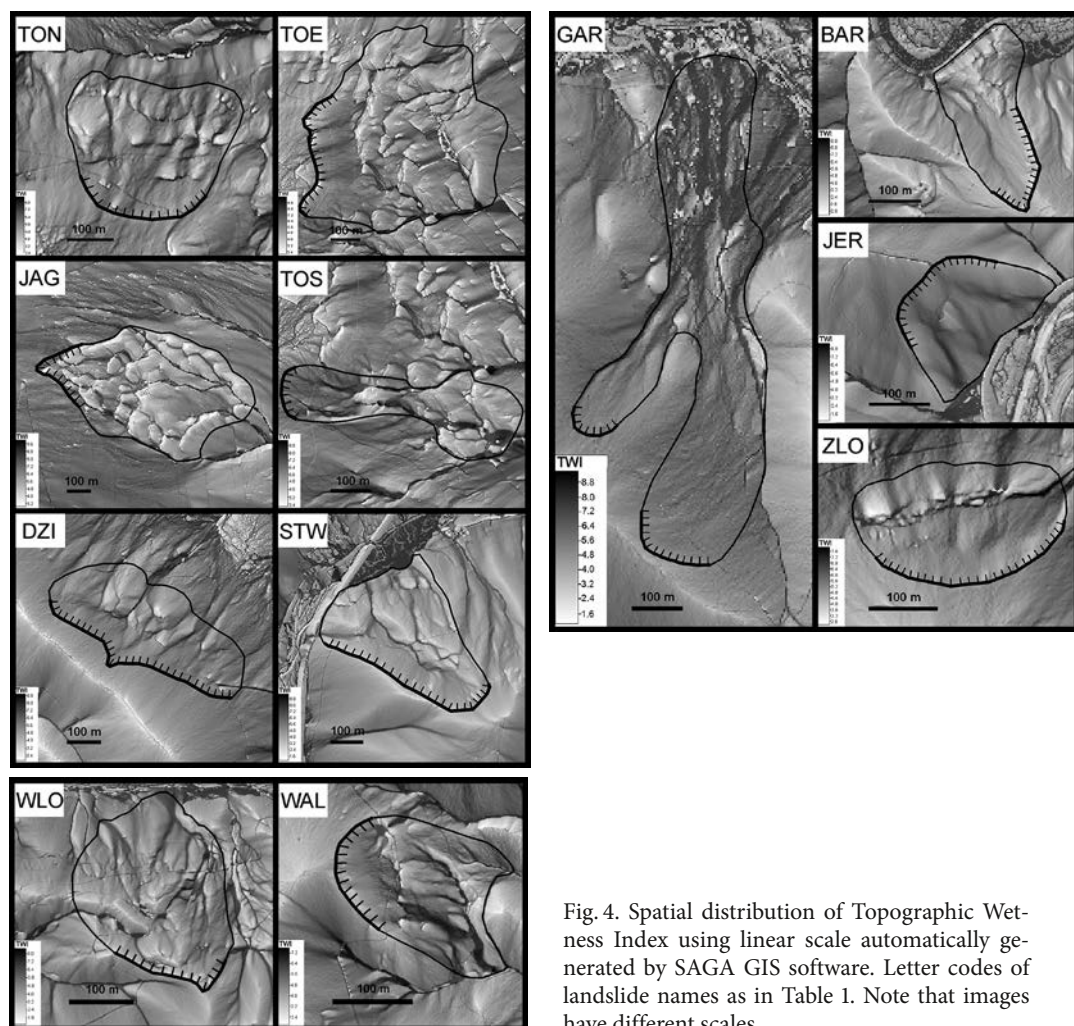


Fig. 4. Spatial distribution of Topographic Wetness Index using linear scale automatically generated by SAGA GIS software. Letter codes of landslide names as in Table 1. Note that images have different scales.

arrangements, resembling braided, dendritic, or apparently random patterns. Examples of a rather simple layout include Bardo and Jerzmanice, whereas Garbatka and Jagodna are characterized by numerous parallel lines, especially in the distal part. Dzikowiec shows a suite of six distinct drainage lines running down the slope. In turn, Toczek E and Włostowa both show areas of concentric and parallel drainage.

In a few cases bipartition of the landslide area emerges (Fig. 6). For example, Toczek E landslide can be divided into the upper part, with slope-perpendicular troughs, and the lower part, where drainage is broadly parallel to the slope. The pattern within Włostowa landslide is analogous, whereas certain similarities can also be found for Waligóra landslide albeit the clarity of slope-perpendicular hollows in the upper part is less clear.

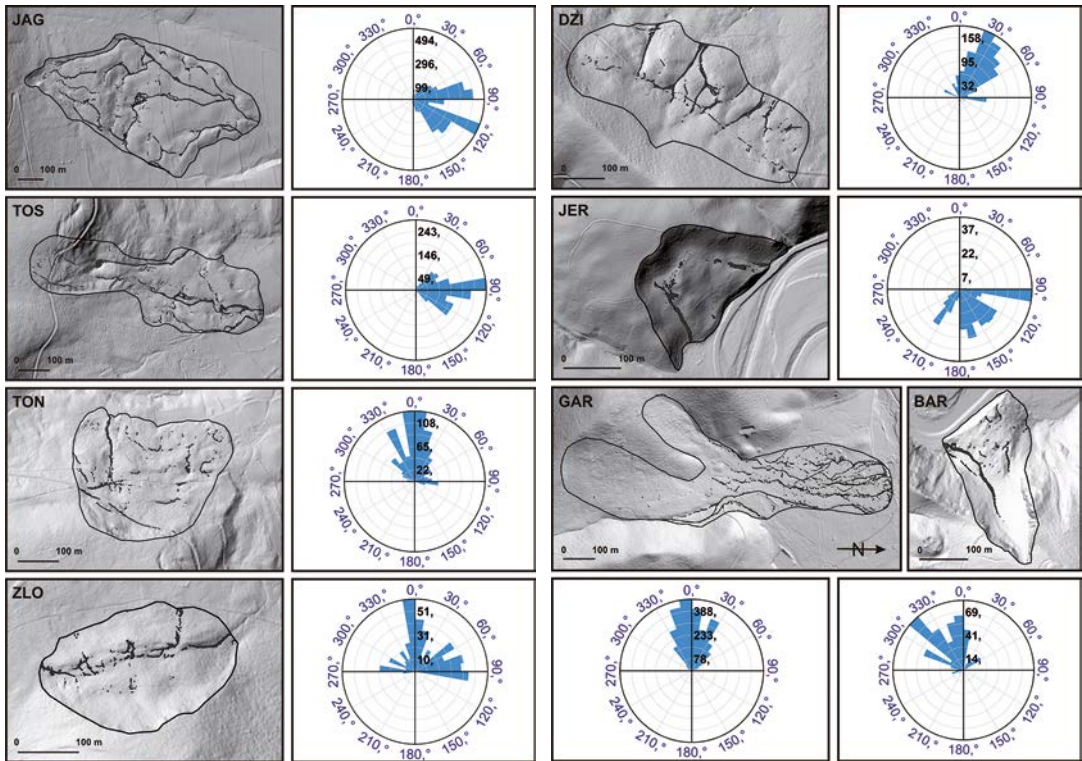


Fig. 5. Reclassified TWI patterns for landslides of relatively simple morphology, showing only very high values, above the mean increased by two standard deviations and rose diagrams to show orientation of linear zones of these values. Circles in rose diagrams indicate cumulative length of segments in metres. Solid black lines indicate the boundaries of each landslide. Letter codes of landslide names as in Table 1.

4.1.2 Directional analysis

Rose diagrams of preferential drainage directions across the landslide body provide further insights into landslide morphology and give the description of TWI patterns a more objective, quantitative dimension (Figs. 5 and 6). A few characteristic layouts can be recognized. Dzikowiec shows one preferred direction straight downslope, consistent with the general slope and shown as a broad (20–40°) maximum on the diagram. Elongated landslide bodies of Toczek S, Garbatka and Jagodna have bi-directional diagrams, with one direction nearly identical with the general slope and another one different by 20–30°. Bardo and Stożek Wielki have bi-directional diagrams too, but here the general slope falls in between these two preferred drainage directions, with one of them dominating over another. Włostowa, Waligóra and Toczek E landslides have multi-directional patterns, with four to five preferred directions, including slope-perpendicular, but none of them is particularly dominant. Złota landslide is a special case, in which the most represented drainage direction expressed as a broad maximum at 70–100° is perpendicular to the slope, although a downslope direction (350–10°) is also represented.

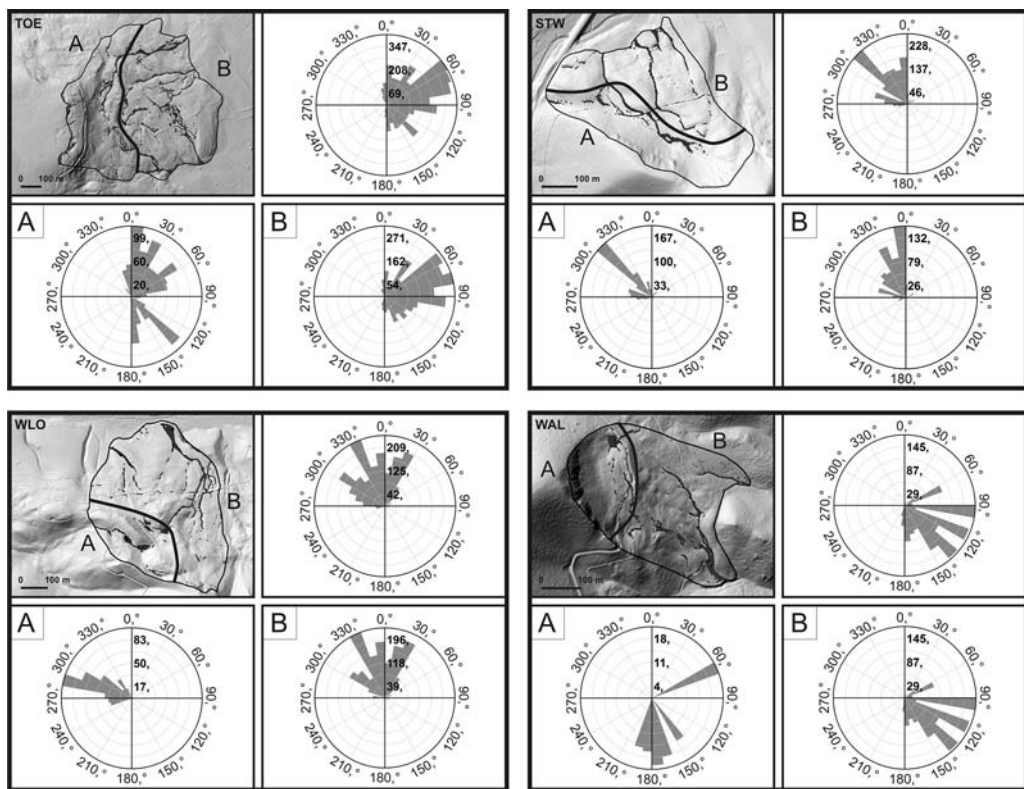


Fig. 6. Reclassified TWI patterns for landslides of distinct bipartite morphology. Only very high values, above the mean increased by two standard deviations are considered. Rose diagrams show orientation of linear zones of these values for the entire landslide, and upper (A) and lower (B) part separately. Circles in rose diagrams indicate cumulative length of segments in metres. Solid black lines indicate the boundaries of each landslide. Letter codes of landslide names as in Table 1.

For landslides of most complex morphology (Toczek E, Włostowa, Waligóra, Stożek Wielki) calculations and rose diagrams have also been prepared for two parts: upper ones, where rigid displaced blocks occur, and lower ones, where considerable surface disruption has taken place (Fig. 6). The resultant diagrams clearly differentiate the landslide body, showing different directions of preferred drainage. Diagrams for the upper sections show directions parallel to the head scarp. The picture is most complex for Toczek E landslide, whereas in the remaining case one (Włostowa, Stożek Wielki) or two (Waligóra) directions emerge. By contrast, diagrams for the lower part show drainage in either perpendicular or oblique (Stożek Wielki) direction to the head scarp, with higher degree of dispersion.

The total length of preferred drainage lines varies from as little as 0.3 km (Jerzmanice) to nearly 4.5 km (Toczek E). This variability is mainly related to the landslide area, but there are three evident and one minor outlier. The three significant ones are Jagodna (152.5 m/ha), Garbatka (136.7 m/ha) and Bardo (133.5 m/ha), whereas the minor one is Toczek S (115.6 m/ha).

Table 2. Quantitative characteristics of TWI and TRI values for landslides.

Landslide name and type	Landslide area (ha)			TWI range			TRI average			Percentage of depositional part of landslide occupied by raster cells with values from the lowest 20/10 % of TRI range	
	Total	Without head scarp	Primary deposition zone	Total	95 % of raster cells	68 % of raster cells	Total landslide area	Total landslide area without head scarp	Primary deposition zone	20 %	10 %
Jagodna Flow	28.3	27.6	27.6	20.13	8.53	4.26	0.13	0.12	0.12	58.96	19.73
Toczek S Flow	13.6	13.6	13.6	18.99	9.73	4.86	0.18	0.18	0.18	58.43	21.91
Toczek E Complex	42.8	37.9	37.0	19.14	7.72	3.86	0.18	0.16	0.16	59.96	21.15
Toczek N Rotational	9.6	7.9	7.2	17.60	7.47	3.74	0.26	0.22	0.22	21.40	4.12
Złota Rotational	4.9	4.0	3.6	16.66	5.48	2.74	0.31	0.24	0.23	9.34	2.78
Dzikowiec Rotational	12.3	11.9	10.4	15.18	5.05	2.52	0.36	0.35	0.33	10.94	2.37
Stożek Wielki Translational	14.1	13.0	11.1	15.93	6.00	3.00	0.39	0.35	0.32	21.27	3.48
Garbatka Flow	21.9	21.7	21.2	19.82	8.35	4.18	0.18	0.18	0.18	48.42	15.76
Włostowa Complex	23.0	22.6	21.9	18.76	7.29	3.64	0.30	0.28	0.28	49.43	9.95
Waligóra Complex	11.8	11.3	10.2	17.79	7.13	3.56	0.31	0.28	0.25	23.52	4.47
Bardo Translational	4.3	3.8	3.5	16.40	6.78	3.38	0.49	0.40	0.38	33.22	6.81
Jerzmanice Rotational	3.5	2.9	2.1	12.11	5.56	2.78	0.36	0.33	0.28	29.89	4.89

4.1.3 *TWI range*

Table 2 shows the range of TWI values for each landslide. It varies from 12.11 (Jerzmanice) to 20.13 (Jagodna), with the mean value of 17.38, which is less than the median. The most elongated landslide bodies (Jagodna, Garbatka) have the highest TWI ranges, but otherwise no clear relationship between landslide morphology and TWI range is possible. Elimination of 5 per cent of raster cells with the highest and the lowest values (> 2 times of standard deviation) improves the clarity of the picture, also resulting in a higher value of coefficient of variation. Elongated bodies retain their leading position, having the highest TWI values (> 8.0), whereas a group of rotational slumps (Jerzmanice, Złota, Dzikowiec) shows the lowest values (< 5.6). Further elimination of 32 per cent of raster cells with the values exceeding one standard deviation does not change the ranking.

4.2 *Terrain Ruggedness Index*

Numerical values of Terrain Ruggedness Index vary by nearly a factor of four (range 0.13–0.49; coefficient of variation 36.84) (Table 2). The highest values are recorded for two translational rockslides (Bardo, Stożek Wielki) which occurred on relatively short and steep slopes, whereas a cluster of low values (0.13–0.18) typifies flow-type displacements or complex landslides with widespread low-lying area experiencing significant surface deformation (Toczek E). Rotational landslides show intermediate values of 0.3–0.36, with those transformed into complex flows in the lower part are characterized by values closer to 0.3 (Włostowa, Waligóra).

The average TRI values calculated for the landslide area without head scarp and for primary depositional zone alone are expectedly lower but the general pattern remains identical. Landslides with the dominant flow component have TRI values below 0.2, rotational ones – in the 0.2–0.3 range (but with two outlying exceptions, each being a fairly atypical case), whilst translational rockslides have TRI values above 0.3. However, reduction of average TRI value reflects the area occupied by the head scarp in the total landslide area rather than its steepness. Most evident decrease was noted for Złota landslide, by more than 22 per cent, but here the head scarp makes nearly 20 per cent of the total landslide area (the highest proportion among all landslides).

TRI values are inversely correlated with TWI range but this correlation seems partly reflect algorithms used to calculate each index. TWI assumes higher values if local slope (expressed as $\tan\beta$) decreases, whereas TRI considers elevation difference within the surrounding cells, hence indirectly slope. The higher this difference, the higher is TRI value.

While analyzing spatial distribution of TRI values special attention was paid to the lowest values within depositional zones of landslides, indicating least local relief. Two variants have been tested, but only results for the first variant, limited to cells whose values were in the lowermost one fifth of the min–max range (i.e. 20 per cent of all values are shown), are presented (Fig. 7).

Landslides subject to this analysis vary considerably in respect to low values of TRI (Table 2). For Złota slump only less than 10 per cent of depositional area falls within the 20 per cent value interval, whereas for Toczek E, Toczek S and Jagodna the respective figures are very close to 60 per cent. In general, considerable disruption of the surface during displacement, typical

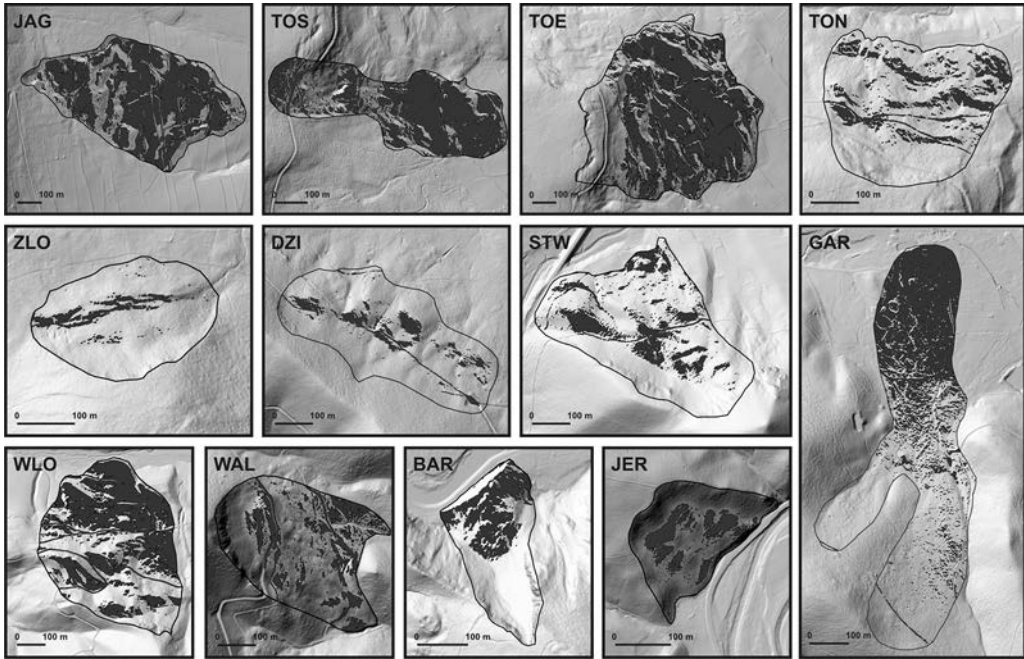


Fig. 7. Spatial distribution of raster cells with values from the lowest 20 per cent of TRI range calculated for the depositional parts of landslides. Solid black lines indicate the boundaries of each landslide. Letter codes as in Table 1. Note that images have different scales.

for flow component of movement, is associated with higher percentages of rasters with low TRI values.

Spatial distribution of low TRI values shows specific patterns, depending on the dominant type of movement. In 'pure' slides linear depressions behind slid blocks are highlighted (Złota, Dzikowiec, Jerzmanice, also Włostowa which was initiated as a series of rotational slumps) and these occur in the upper parts of landslide bodies. By contrast, for flow displacements low TRI values typify subdued, distal parts (Toczek S, Garbatka). Rather unusual patterns have been obtained for Jagodna and Toczek E, where the low values are widely distributed over the entire landslide bodies.

5 Interpretation and evaluation

Results of our analyses generally show that both TWI and TRI computed from high-resolution digital elevation data perform well and provide information which may be helpful in landslide characterization. Although the respective algorithms to calculate these indices both include, directly or indirectly, local slope in the surrounding of a raster cell, information potential of each index is different, especially if spatial patterns are considered.

Ranked numerical values of TWI and TRI, expressed as TWI range within a landslide and TRI average for a landslide, respectively, reveal certain relationships with the dominant type of

displacement (Table 2). TWI range ranking, especially the one disregarding 5 per cent of rasters with the most extreme values, clearly differentiates displacements dominated by flow and resulting in considerable surface disruption, from slumps, in which failed mass retains coherence and rigidity. The former are characterized by TWI range values above 8, whereas the latter show values below 6. Complex landslides, involving both slide and flow, have intermediate values, between 7 and 8, whereas translational rockslides fall in the 6–7 interval. Similar, although not identical relationships emerge from TRI ranking. Here, flow-dominated displacements have values below 0.2, while the highest values typify translational rockslides. Displacements with dominant rotation assume intermediate values, and the more surface disruption due to subsequent transformation into flow, the lower is TRI value.

These relationships mean that displacements resulting in significant disruption of original morphology tend to produce topographic surfaces less diversified in terms of elevation difference. However, visual inspection of digital terrain models (DTMs) shows that these surfaces may be characterized by considerable local roughness (i.e. change of slope and aspect over small distances), consistent with rheology of flow displacement (DE BLASIO 2011) and common emphasis on hummocky or highly irregular morphology produced by cohesive flows (e.g. ZARUBA & MENCL 1982, MCKEAN & ROERING 2004, CROZIER 2010). Consequently, one might also expect higher drainage density for flow-dominated displacements and this emerges from our dataset, in which drainage line densities for flows are higher than for other landslides. However, no differentiation between slumps and complex landslides can be made using this parameter.

Spatial pattern of TWI distribution for landslides of different types shows certain specific features. Visual analysis alone indicates that drainage can be distributed across the landslide area or concentrated along a few or even one main line. The former case typifies wide unconfined flows (Jagodna, Garbatka in the lowermost part), while translational wedge rockslides tend to have one drainage line parallel to the head scarp. While in most instances drainage is heading downslope, rotational slumps may have distinct ‘wet’ areas perpendicular to the general slope, corresponding with linear depressions behind back-tilted slid blocks. This pattern is best seen for Złota and the upper part of Włostowa landslide. Such depressions are not present at Dzikowiec landslide, despite its rotational nature. However, its morphology suggests that mass movements occurred as several disconnected slides rather than one large slump. Consequently, drainage of the head scarp area occurs via a series of downslope-oriented troughs separating these minor slid blocks.

Visual analysis can be enhanced by a more quantitative one, focused on directional characteristics of drainage lines. While this approach is unable to reveal the presence of several parallel lines of identical orientation, both preferred directions of drainage and their relation to the general slope can be shown. Although our sample is small, rose diagrams show a variety of drainage types, ranging from distinctively uni-directional to multi-directional, with more directions being typical for landslides of complex morphology. However, our initial assumption that rotational slides can be easily separated from other ones due to the strong signal from a trench parallel to the head scarp, was not confirmed. Only one slide, Złota, shows this pattern and the reason is that the slid block has undergone minimal fragmentation while moving downslope, hence drainage tends to be diverted laterally. Otherwise, drainage is being or-

ganized in between disintegrated blocks and downslope directions appear in high proportion. However, splitting large complex landslides in two parts has resulted in two very distinctive and different patterns for the upper, rigid part and the lower, disrupted part. Geometry of the head scarp, reflected in its concavity in plan, plays a role too, as exemplified by Złota and Jerzmanice slides. Both are rotational slumps, but the head scarp of the latter is much more arcuate. The resultant drainage along the lateral sections of the main scarp gives a dominant signal in directional frequency analysis.

Both TWI and TRI may be computed and presented with certain modifications imposed by the operator. We tested several variants, with the following findings. First, elimination of extreme values of TWI and consideration of 95 per cent of rasters resulted in a ranking that fits better the typology of landslides inferred from the recognition of their morphological features. Second, analysis of spatial distribution of rasters with very high values helps to locate preferred drainage lines and their patterns appear specific for different types of landslides. Third, computation of average TRI values may be carried out including or excluding the head scarp. In the latter variant, morphological diversity of landslide body is better shown and separation of flow displacements from landslides typified by sliding of rigid rock masses is more evident.

6 *Recommendations and conclusions*

Interest in objective presentation of topographic complexity of landslide surfaces by means of geomorphometry is not new. However, it was typically driven by either willingness to improve landslide identification on digital elevation models (McKEAN & ROERING 2004, VAN DEN EECKHAUT et al. 2012) or, less often, served the purpose of distinguishing landslides of different ages (GLENN et al. 2006, LAHUSEN et al. 2016). Various measures of surface roughness have been used, from simple to rather complicated ones, although BERTI et al. (2013), after comparing the performance of many of them, concluded that both types of indices work fairly well.

Our focus in this paper was different and we intended to check if Terrain Ruggedness Index (RILEY et al. 1999), used in landslide studies before, and Topographic Wetness Index (GRUBER & PECKHAM 2009), hardly used so far, can be of added value in geomorphic characterization of landslides. The latter was of particular interest and we assumed that distribution and clarity of lines of preferential drainage may yield significant information about landslide morphology and, by inference, origin, otherwise unaccounted for.

The main conclusions from our methodical experiment, carried out on elevation data derived from high-resolution LiDAR are the following. Both TWI and TRI proved capable of differentiating landslide population into smaller groups, consistent with origin and mechanism of displacement. Thus, translational rockslides, rotational slumps, flows, and complex features have distinct geomorphometric signatures as far as these two indices are concerned. TWI provides additional information about relief diversity within the landslide body, indicating patterns of preferential drainage in both qualitative and quantitative way. Landslides of different types show different spatial distributions of TWI. TRI average values for entire landslide areas are strongly influenced by the presence of steep scarp, whereas TRI analysis limited to the lowest values revealed considerable differences between landslide bodies resulting from slide, with

little local surface disruption, and flow, with extensive surface disruption. The latter are much more subdued and TRI shows this characteristics fairly well.

Topographic Wetness Index may be analysed in different ways. Visual analysis of TWI spatial distribution is attractive and helps to identify surface and drainage complexity. However, due to size of datasets and considerable range of TWI values within rough terrains colour scales typically used for visualization (especially if linear scale is adopted) do not return TWI diversity with sufficient clarity and make identification of raster cells with the highest values difficult. Therefore, lines of most preferred drainage are best shown if only extreme values separated from the large dataset through reclassification are considered. On the other hand, the sample is better diversified, broadly consistent with landslide morphological diversity, if these extreme values are ignored and only 95 per cent of rasters are subject to statistical analysis. Directional analysis of lines of preferential drainage within the landslide bodies shows that different types of displacement have their specific morphological signatures. In conclusion, we argue that both TWI and TRI do provide new information about landslide morphology and their potential in landform studies is worth further exploration, not limited to landslides.

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